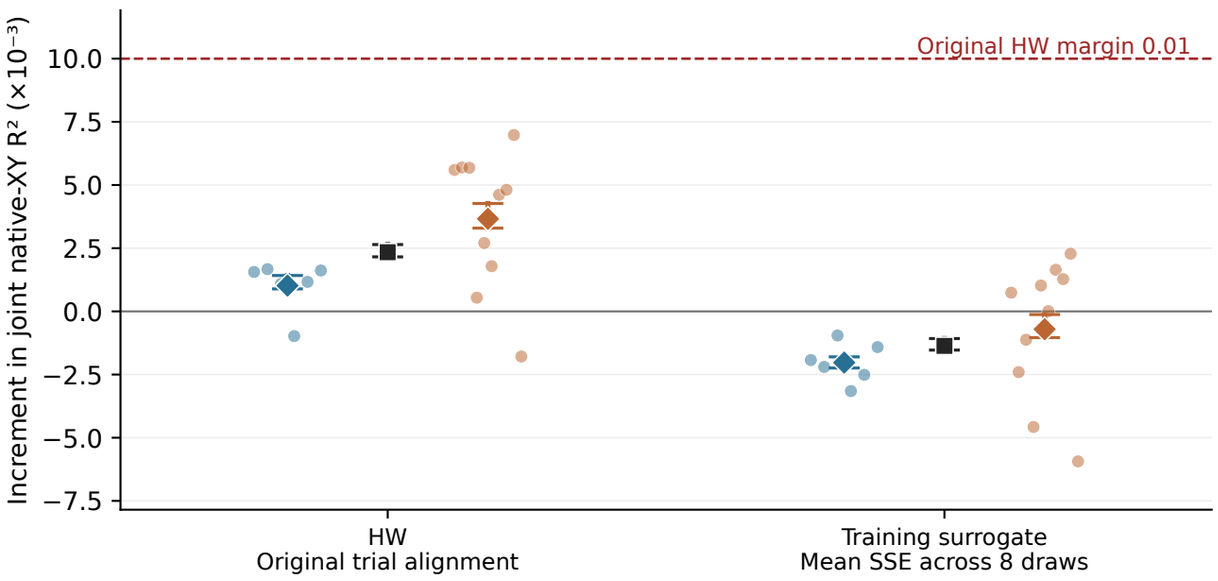
HW retains an advantage over capacity-matched training surrogates

All 16 paired effects are positive; HW primary gain 0.002341 still falls below 0.010.

Exact shared Gram and ridge df in every fold · selected λ=1 · df 46.124–73.894 across 128 direction folds

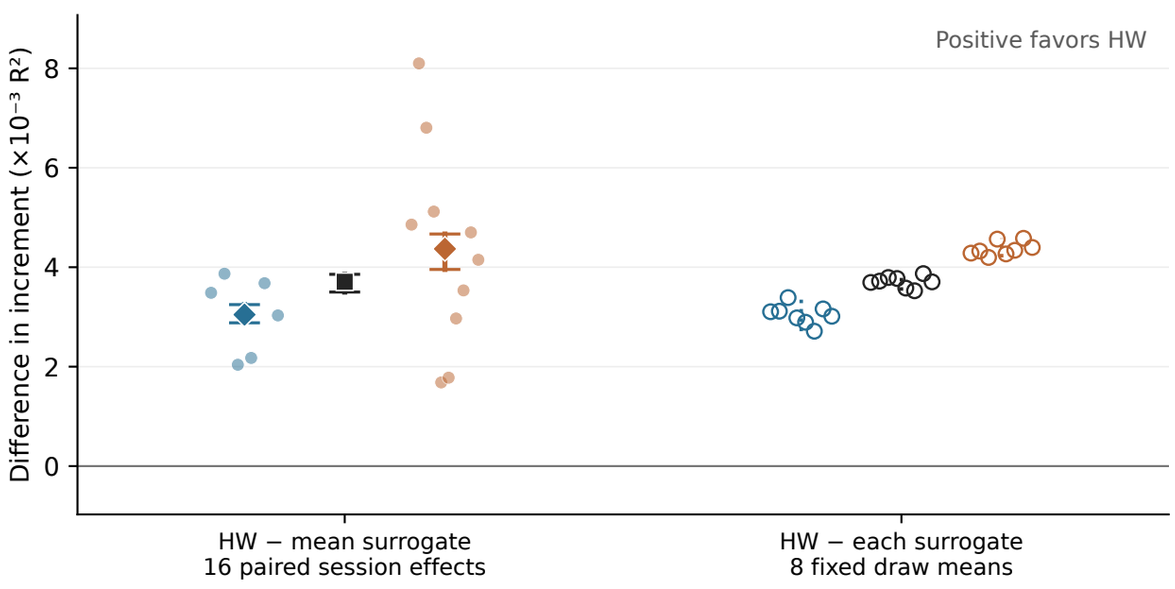
A Prediction gain over direction/time baseline



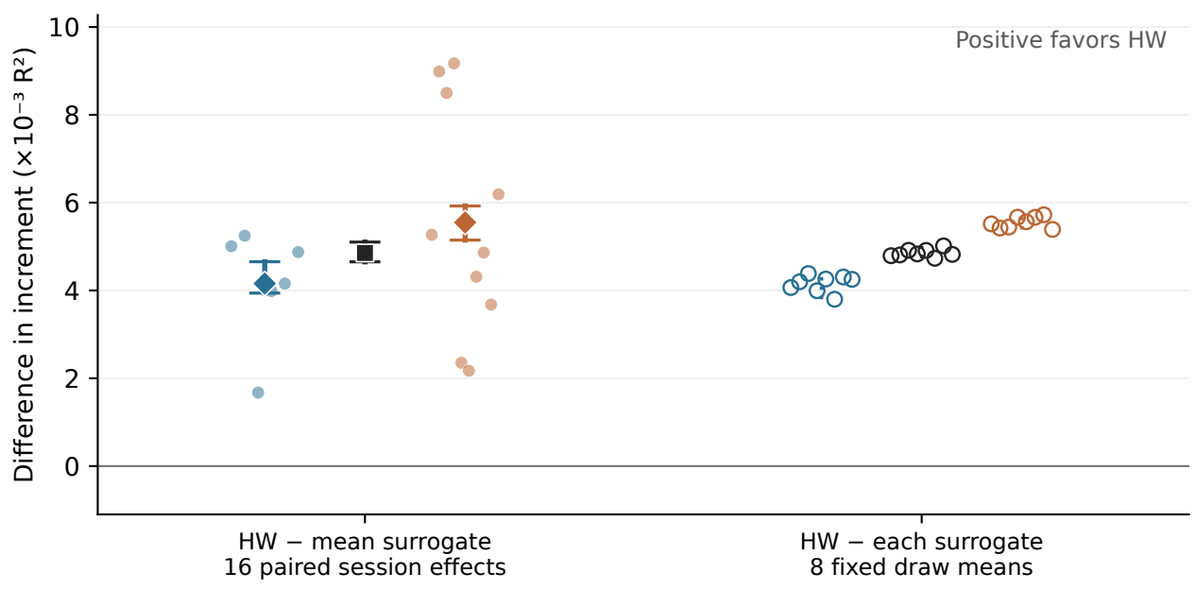
B Correct versus wrong within-direction trial pairing



C Direct prediction advantage: HW minus surrogate



D Trial-pairing advantage: HW minus surrogate



◆ Mihili mean ◆ Chewie mean ■ Equal-animal mean ○ Individual draw mean

Filled small dots: all 16 sessions. Solid bars: fixed-prediction session-omission ranges. Open dots/dotted spans: all 8 draw means and their range.
These descriptive ranges are not confidence intervals or p-values; the 8 draws are finite controls, not independent biological replicates.
Whole training-trial feature histories are permuted within direction. Time, direction structure, preprocessing, B, targets and selected λ remain fixed.
Original HW is never refit. Evaluation features stay paired with their original trials; identity scores separately wrong-pair held-out trajectories.
Surrogate correct/wrong-pair SSEs are averaged, not predictions. Direct contrasts share the original native-XY denominator. Every panel retains all signed values.
This is a conditional capacity control, not a signal-free or raw-phase null. Exposed development data · 2 animals · original primary failure remains.